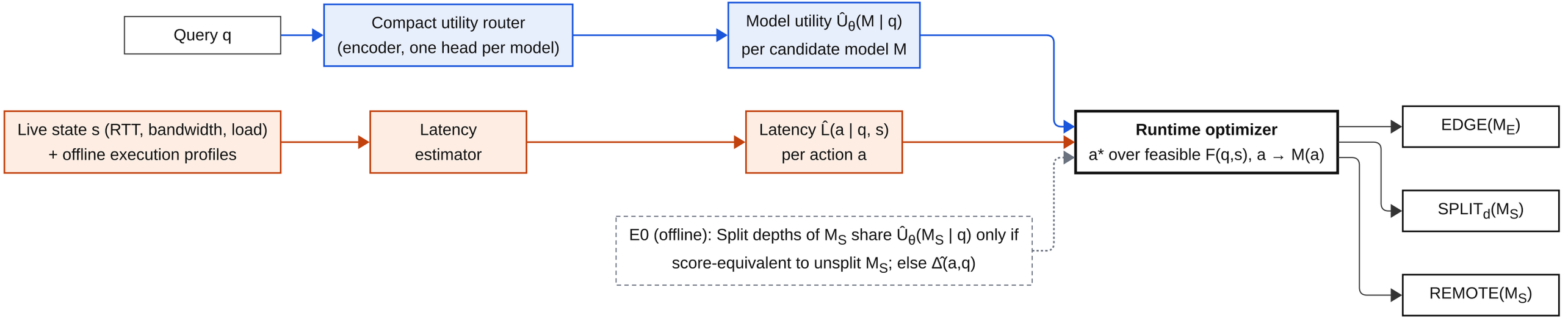
Query q
Compact utility router
(encoder, one head per model)
Model utility $\hat{U}_\theta(M \mid q)$
per candidate model M
Live state s (RTT, bandwidth, load)
+ offline execution profiles
Latency
estimator
Latency $\hat{L}(a \mid q, s)$
per action a
Runtime optimizer
a* over feasible F(q,s), a → M(a)
EDGE(M_E)
$SPLIT_d(M_S)$
REMOTE(M_S)
E0 (offline): Split depths of M_S share $\hat{U}_\theta(M_S \mid q)$ only if
score-equivalent to unsplit M_S; else Δ(a,q)